Aarush Mane

Robbinsville, NJ | am6927@princeton.edu | (609) 610-4280 | Portfolio | LinkedIn | GitHub

Education

Princeton University Princeton, NJ
B.S.E., Computer Science | Prospective Minors: Mathematics, Robotics *Expected May 2030*

- **Activities:** PAVE (Autonomous Vehicle Engineering); Princeton Robotics Club; South Asian Students Association (SASA)

Research

NJ Governor's School of Engineering & Technology July 2025
Research Scholar | IEEE MIT URTC 2025 Co-author *Rutgers University, NJ*

- Developed Python PID control for TurtleBot3 differential-drive navigation, using moving-window integral, velocity-based yaw targeting, anti-windup, and motor saturation; validated performance in hardware trials.
- Compared A* and Dijkstra path-planning runtimes ($p < 0.01$); evaluated obstacle avoidance and waypoint tracking with MATLAB, Python, Gazebo, LiDAR-equipped TurtleBot3 hardware, and Vicon motion capture.
- Presented to 300+ attendees at GSET; co-authored *PID Control Optimization and Trajectory Tracking for Autonomous Differential Drive Robots* (IEEE MIT URTC 2025).

Technical Experience

FIRST Robotics Competition Team 2590, Nemesis Oct. 2023 – June 2026
Software Development & Strategy Lead *Robbinsville, NJ*

- Developed Java/OpenCV AprilTag PnP pose-estimation pipelines with OV9281 cameras; used checkerboard calibration, fisheye distortion correction, and exposure tuning; prototyped Oculus Quest localization to reduce odometry drift.
- Reproduced OrangePi vision dropouts on a battery-powered testboard, traced intermittent pose updates to inadequate supply current, and restored stable camera measurements by correcting coprocessor power delivery.
- Wrote Java command wrappers for JSON motion profiles and autonomous subsystem coordination; analyzed Excel/Sheets scouting and alliance data for a 65-member team placing 5th at the 2024 FIRST World Championship.

FIRST Tech Challenge Team 22348, REAPR Oct. 2022 – June 2026
Founder & Software Mentor *Robbinsville, NJ*

- Founded a 15-member FTC team; raised $16,000+ in sponsorships and increased local robotics participation 30%.
- Trained TensorFlow object detection on tens of thousands of images; tuned OpenCV preprocessing, bounding boxes, exposure, and detection thresholds for autonomous targeting.
- Programmed Java mecanum-drive kinematics, odometry-based trajectory planning, and autonomous routines; built an AppSheet scouting app to evaluate opponents and shape match strategy.

Projects

Nemo – AI-Assisted Ocean Waste Source Tracing | GitHub | Live demo Oct. 2026

- **Awards:** 3rd Best Overall Hack; 2nd Place, ASUS "Save the Earth" Sustainability Track.
- Developed Python regression tests for JSON experiment exports, catching missing, duplicate, contradictory, and false-positive records; verified SHA-256 hashes, provenance, and data integrity across repeatable trial exports.
- Designed Grok-assisted transport-model instructions for advection–diffusion, RK4 numerical integration, and Kalman state estimation; proposed repeatable simulation trials, baseline comparisons, and held-out evaluations.
- Specified blind experiment and evidence-tracking protocols separating tank measurements, synthetic simulations, and archived ocean data; presented Nemo to HackHarvard judges and sponsors including Capital One.

Argus – Robotics Team AI Assistant Apr. 2026 – May 2026

- Built a multi-user Slack assistant linked to GitHub code and scouting records, enabling teammates to inspect commits, trace Java robot failures, and coordinate debugging during time-sensitive competitions.
- Automated match-log and scouting ingestion with Python REST APIs and GitHub Actions; converted competition records into visual analyses of team performance, matchups, and alliance-selection tradeoffs.

Additional Experience

iStart Valley Jan. 2025 – June 2025
Financial Projections & Business Strategy Intern *Remote*

- Modeled BioPack's costs, margins, pricing, revenue forecasts, and market for edible and biodegradable packaging.
- Developed a go-to-market strategy with a proposed 10–20-business pilot; pitched BioPack to experts, winning 1st in the cohort, reaching Global Semifinalist, and receiving a $500,000 valuation.

Technical Skills

Programming & Tools: Java, Python, MATLAB, HTML/CSS, Git, GitHub Actions, JSON, REST/Slack APIs, Linux, Docker, Excel, Google Sheets, AppSheet

Robotics & Modeling: OpenCV, TensorFlow, ROS 2, Gazebo, LiDAR, Vicon, AprilTags, PnP, camera calibration, pose estimation, PID, kinematics, A*, Dijkstra, RK4, Kalman filtering